

3	(a)	The spring <u>does not obey</u> Hooke's Law. The straight line in the graph <u>does not pass through the origin</u> OR The tension <u>force</u> in the spring is <u>not proportional to the extension</u>. * Answer must derive evidence from the graph.	1
	(b)	Remove the masses and the <u>spring should return to its original length</u> if the spring is not permanently deformed.	1
	(c)	Work done is <u>obtained from the area under the graph multiplied by g</u>. * Student must attempt to explain the approach taken. Work done = $\frac{1}{2} [(70 + 145) \times 10^{-3}] \times 120 \times 10^{-3} \times 9.81$ = 0.127 J (0.13 J)	1
4	(a)	By the principle of conservation of energy	